

Master's Programme in Building Technology

Integrating AI into Procurement Processes in Construction

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**Master's thesis
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Abstract

The objective of this research is to identify opportunities for utilizing artificial intelligence in different phases of the procurement process and to explore its possible integration into them. Potential challenges and biases regarding the integration were identified, and strategies to mitigate them were explored. A case-study was conducted to evaluate how generative AI can be used to enhance bid comparison, which led to the development of a customized GenAI prompt for Skanska to utilize in bid comparison.

The study explores the set objectives by conducting a comprehensive literature review, interviews with five procurement professionals, and a case-study. Empirical research section of this thesis was done in two parts. First, the interviews explored the current procurement process and its challenges, followed by a discussion on the potential usage of AI-based tools in this process. Second, the case study measured how generative artificial intelligence (GenAI) can be utilized to improve the accuracy and efficiency of bid comparison in procurement. The case-study employed knowledge obtained from previously conducted literature review and interviews.

The research identified four key procurement phases where AI could potentially provide the most value in: 1) bid comparison, 2) supplier evaluation, 3) procurement package formation, and 4) contract oversight. The case-study tested Skanska's own GenAI tool with a generated prompt to conduct bid comparison from pre-linked bid files. The results were compared to a manually performed one, and the results revealed inconsistencies with GenAI's performance but also potential for a GenAI-based tool to enhance bid comparison's efficiency. GenAI's capability to provide textual analyses from the linked files was determined as the first step on GenAI's integration to the process of bid comparison.

This research offers concrete data on AI's potential in procurement processes. Future development options are addressed, which consists of training the GenAI tool to compare more complex offers more accurately and to start looking into possibilities on how AI could enhance the three other procurement phases where it was deemed to possibly provide value in.

Keywords GenAI, Procurement, Bid comparison, Digitalization

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Tiivistelmä

Tässä työssä tutkittiin tekoälyn hyödyntämismahdollisuuksia hankintaprosessin eri vaiheissa sekä sen integrointia osaksi näitä prosesseja. Tavoitteena oli myös tunnistaa tekoälyn käyttöönoton haasteet ja ennakkoluulot, sekä esittää keinoja niiden vähentämiseksi. Lisäksi toteutettiin tapaustutkimus, jossa arvioitiin generatiivisen tekoälyn (GenAI) käyttöä tarjousten vertailun tukena. Skanskalle kehitettiin tätä varten räätälöity GenAI-kehote tarjousvertailua varten.

Asetettuja tavoitteita tutkittiin laajan kirjallisuuskatsauksen, viiden hankinnan ammattilaisen haastattelujen sekä tapaustutkimuksen avulla. Empiirisen tutkimuksen osa toteutettiin kahdessa osassa. Ensimmäiseksi, haastatteluissa kartoitettiin nykyistä hankintaprosessia ja sen haasteita, sekä selvitettiin tekoälypohjaisten työkalujen potentiaalista käyttöä hankintaprosesseissa. Toiseksi, tapaustutkimuksessa mitattiin miten generatiivista tekoälyä (GenAI) voidaan hyödyntää parantamaan tarjousten vertailun tarkkuutta ja tehokuutta. Tapaustutkimus hyödynsi edukseen tietoa, jota oltiin hankittu ennalta toteutetuissa kirjallisuuskatsauksessa ja haastatteluissa.

Tutkimuksessa tunnistettiin neljä hankinnan vaihetta, joihin tekoäly voisi antaa eniten lisäarvoa: 1) tarjousten vertailu, 2) toimittajien arviointi, 3) hankintapakettien koostaminen, ja 4) sopimusvalvonta. Tapaustutkimuksessa testattiin Skanskan GenAI-työkalua luodun kehotteen avulla suorittamalla tarjousvertailu ennalta linkitetyistä tarjousasiakirjoista. Vertailun tuloksia verrattiin manuaalisesti tehtyyn vertailuun. Kyseisistä tuloksista ilmeni sekä GenAI:n suoritustekyvyn liittyviä epäjohtomukaisuuksia, että potentiaalia GenAI-pohjaisen työkalun hyödyntämiseksi tarjousvertailujen tehostamisessa. Työkalun kyky tuottaa tekstimuotoisia analyysejä linkitetyistä tiedostoista todettiin ensimmäiseksi askeleeksi GenAI:n integroimisessa tarjousvertailuprosessiin.

Tutkimus tarjoaa konkreettista dataa tekoälyn potentiaalisesta käytöstä hankintaprosesseissa. Työssä tunnistetaan tulevia kehitysmahdollisuuksia, jotka sisältävät GenAI-työkalun kehittämisen monimutkaisempien tarjousten tarkempaan vertailuun sekä mahdollisuuksien selvittämisen tekoälyn hyödyntämisestä kolmessa muissa tunnistetuissa hankintaprosessin vaiheissa.

Avainsanat Generatiivinen tekoäly, Hankinta, Tarjousvertailu, Digitalisaatio

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Otaniemi, 18 May 2025
Joni Hamppi

1 Introduction

1.1 Background

The construction industry has historically been characterized as a low performer among innovation-driven sectors with limited improvement in productivity and growth. The productivity growth has averaged only 1% annually over the last twenty years, raising questions about the construction industry's efficiency (Regona et al., 2022). The deficit in growth is due to the construction sector's culture of resistance to change and perseverance to old habits (Abioye et al., 2021). To keep pace with ongoing development, innovative solutions are required to address emerging challenges. This is why the industry is increasingly adopting digital and intelligent methods, including artificial intelligence (AI), to address these challenges. (Salehi & Burgueño, 2018).

This thesis focuses on utilization of AI in procurement processes. Procurement is an active process in construction projects throughout their entire lifecycles. Its processes contribute broadly to construction project costs and their management. They include the preparation of bids, supplier selection, and contract administration. These processes impact project costs, quality, and timelines. Materials used in a construction project can represent as much as 70% of the project's total costs (Donyavi et al., 2024). This calls for improvement and a system that can systemically reduce the costs linked to procurement processes in construction projects so that the overall project costs stay as low as possible.

Regarding procurement processes and systems, people have a habit of choosing a procurement procedure they have prior knowledge of (Eriksson & Westerberg, 2011). To successfully introduce a new procurement procedure, workers must be comfortable with its application and see positively its potential impact on project outcomes (Tysseland, 2008). With the help of AI, there is potential to enhance the efficiency of procurement processes in the

construction industry. Ideally, AI would act as a supporting tool that enables decision-making to be faster than before. This would release resources to other tasks faster and smoothen operations amongst experts.

Most commonly, AI is defined as the ability of a digital computer or computer-controller robot to perform tasks. Its roots come from the 1950s when the first computer was invented (Chen & Ying, 2022). Nowadays AI has a substantial subfield where machine learning (ML), natural language processing, and robotics have been applied to address complex challenges and assist in making decisions for industry related issues. (Abioye et al., 2021). Other fields have received notable improvements in service processes, business operations, and industry productivity in recent years. Despite the improvements in these fields, construction industry is still in the early stages of realizing the full potential of AI in its processes (Abioye et al., 2021).

The emergence of digitalization has not increased the productivity of Finnish construction sector in recent decades. This is because the sector suffers from conservatism (Seppänen, 2022). New solutions are not adopted because they would require changes to existing operating practices and business models (Seppänen, 2022). The reluctance to adopt digital tools to everyday processes prevents the construction industry from improving its productivity, design and quality (Hossan & Nadeem, 2019).

1.2 Objective and Research Questions

This research aims to identify opportunities for utilizing artificial intelligence in different phases of the procurement process and to explore its possible integration into them. Through the integration of AI into these processes, there could be potential to improve the reliability, efficiency, and overall effectiveness of procurement in the construction industry.

A key aspect of this research is to examine the bid comparison phase of the procurement process and explore how GenAI could enhance its efficiency. This phase was selected for more detailed analysis based on Skanska Talon-rakennus Oy's interest in gaining a deeper understanding of its current practices and potential for improvement. This is because an effective bid analysis is critical, as decisions based on submitted bids can have significant cost implications for a project. Additionally, poor analysis of bids can include overlooking contractor reliability or past performance.

By leveraging AI-assisted tools, there is potential to minimize these risks, which would enable more efficient and accurate bid evaluations. Also, a structured AI-driven analysis facilitates informed decision-making, providing stakeholders with clearer insights and potentially reducing costs.

The research is structured around the following questions:

1. What opportunities does artificial intelligence (AI) in general offer in different phases of the procurement process?
 - What kind of improvements can be achieved through its utilization to productivity, cost, and timelines?
2. What potential challenges and biases need to be identified and managed when integrating AI into procurement workflows, without disrupting existing processes during the integration phase?
3. How can generative artificial intelligence (GenAI) be utilized to improve the accuracy and efficiency of bid comparison in procurement?

The first research question aims to point out successes, challenges, and areas for improvement in current procurement processes as a whole. By understanding the current status of the procurement process, the potential utilization of AI and its improvement possibilities to the processes can be identified and presented. The second research question aims to identify potential

challenges and biases that may arise when integrating AI-based applications to company processes. By gaining an understanding of these challenges and biases, proper approach to AI integration can be achieved. Lastly, the final research question aims to study how GenAI could be utilized in procurement related bid comparison. The goal is to measure the accuracy and efficiency of GenAI-based bid comparison.

This thesis has been commissioned by Skanska Talonrakennus Oy and it focuses on the development of its internal procurement processes. Based on these research questions, it is paramount to understand Skanska's procurement process and find opportunities that align with them.

The objectives of this research are to:

1. Identify opportunities for AI integration in different phases of the procurement process and evaluate the improvements they can deliver.
2. Explore possible integration strategies for AI into procurement processes by identifying prejudices and concerns related to it.
3. Develop and test a customized GenAI prompt within Skanska's own AI system to enhance bid comparison's accuracy and efficiency.

1.3 Scope of the Thesis

This thesis explores the integration of artificial intelligence (AI) into the procurement processes conducted by the main contractor at Skanska Talonrakennus Oy. The study focuses on identifying opportunities for AI implementation across different stages of the procurement process and assessing the operational improvements it can bring. It also examines strategies for integrating AI into existing workflows by also addressing potential challenges and concerns.

A key focus of this research is understanding Skanska's procurement process and identifying areas where AI can be effectively applied. This study will explore more deeply how generative AI can be used to enhance bid comparison, leading to the development of a customized GenAI prompt for Skanska to boost accuracy and efficiency of bid evaluations.

This research will specifically define the stage within the procurement process at which bid comparison begins. Other areas of the procurement process will be excluded from AI-based testing at the request of Skanska Talonrakennus Oy. The scope is justified by the multiphase nature of procurement which encases an extensive amount of information and theory. To ensure the quality and reliability of this research, GenAI testing will be conducted solely on the bid comparison phase.

Machine learning and deep learning are mentioned in this research, but their further analysis will not be addressed. Generative artificial intelligence will be researched and all the findings in this research will be based on its applications. This can be naturally explained by the emergence of GenAI tools in people's everyday lives, along with their accessibility and user-friendly platforms.

The research will neglect strategic procurement and focus on operational procurement given the construction industry's project-specific procurement needs. This is because strategic procurement refers to long-term procurement goals at the corporate and company level whereas operational procurement refers to day-to-day procurement activities that support ongoing operations and projects of a company, such as a construction project. (Unite, 2024). According to my experience and my advisor's description, construction projects often involve numerous material and subcontracting purchases under tight schedules. This is why operational procurement offers a more

relevant and practical context for researching the potential of integrating AI into procurement processes in construction.

1.4 Research Methods

This research is a multiphase project that will aim to enhance procurement processes in construction. Multiple research methods will be utilized to achieve the set objectives focusing on both theoretical and empirical approaches.

1. Literature review

This research starts with a literature review (chapters 2-4) that examines scientific studies on procurement in construction companies and the use of artificial intelligence in the construction industry. The literature review will act as a foundation for the rest of the study, and it aims to:

- Define current procurement process and practices in the construction industry with also reviewing digitalization.
- Identify potential challenges in procurement.
- Explore how AI can potentially enhance construction industry's productivity and procurement process.

2. Professional interviews

The literature review will be followed up with qualitative interviews where five construction industry procurement professionals will be interviewed. The interviews will be divided into two semi-structured parts.

The first part will provide insight into:

- The current state of procurement process in construction.

- The existing and potential role of AI in different phases of the procurement process, and its potential to deliver improvements.
- Areas where AI could potentially enhance procurement processes within the company.
- Challenges and biases within workers when integrating AI into procurement processes.

The second part of the questions will offer expert's opinions on how AI could be integrated into the process of bid comparison and how its performance could be enhanced.

The findings discovered in these interviews will be presented and development proposals will be presented for future research.

3. A case study of GenAI in bid comparison

A case study will be conducted to test concretely how GenAI can enhance bid comparison at Skanska Talonrakennus Oy. This phase consists of:

- Developing a customized prompt for Skanska's own GenAI system. The prompt will be based on interview insights, Skanska Talonrakennus Oy's guidelines, and scientific literature.
- Testing the GenAI-system to analyse and compare bids from 6 different comparable subcontractors.
- Comparing the results of both the GenAI-system and a person to assess the accuracy, efficiency, reliability, and potential of the outputs for integration into bid comparison.

This case study will provide insight into the potential of Skanska's own GenAI tool usage in comparing bids and its potential for integration into Skanska Talonrakennus Oy's procurement workflows.

2 Procurement in Construction

2.1 Different Types of Procurement in Construction

Procurement processes in construction vary from project to project. A suitable process for an individual project depends on the type of construction, availability of resources, client experience and volume of work to be done (Sani Bako, 2016). The share of material procurement and subcontracting typically accounts for 60–80 percent of a project's total costs. Therefore, it is accurate to note that the success of procurement has a significant impact on the project's overall financial and schedule outcomes (Junnonen & Kankainen, 2012). The substantial proportion of procurement in total project costs encourages companies to obtain all services and materials not solely based on the lowest price, but based on the best overall value they can offer.

Procurement is the process of creating, fulfilling, and managing contracts associated with the supply of goods, services, disposals or construction works, or any combination of these. It is a critical component when planning and executing construction projects because companies often rely on other companies' goods and services to reach their goals (Watermeyer, 2012). Procurement can also be characterized as the management of an organization's external resources (Junnonen & Kankainen, 2012).

The largest share of material in procurements is in construction product purchases, whereas it is lowest in service purchases. This illustration can be seen in Figure 1 below (Junnonen & Kankainen, 2012).

In construction, procurements can be grouped into the acquisition of construction products, subcontracts, and services, as shown in figure 2 (Junnonen & Kankainen, 2012). Figure 2 presents a classification model for construction procurement. The model is based on two dimensions, the nature of the procurement (standardized vs. project-specific) and the type of

procured item or service. The contract type shows what kind of contract form is appropriate for each type of item or service.

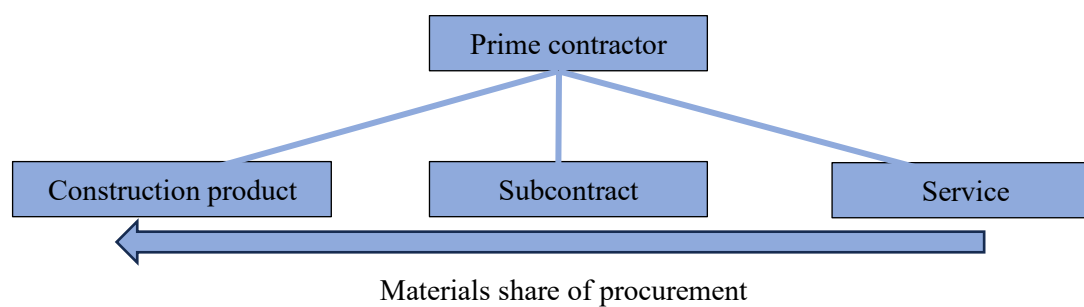


Figure 1: Materials share of procurement (Junnonen & Kankainen, 2012).

Procurement type		Contract type
Standardized	Project specific	
Construction product	Project-specific construction product	Procurement contract
Small procurement		
Subcontract	Project-specific subcontract	Subcontract agreement
Service	Project-specific service	Rental agreement, consulting agreement, design agreement, transportation agreement

Figure 2. Compartmentalization of procurements (Junnonen & Kankainen, 2012).

Construction companies have multiple ways and processes to procure materials and services from a supplier. It is usually dependent on the scale of the procurement and what kind of procurement type is used. The most commonly used methods include single-source procurement, multiple-source procurement, two-stage tendering, open tendering, selective tendering, and negotiated tendering. Each of these methods have their own characteristics, advantages, and suitable applications.

Single-source procurement

A construction company procuring all its services and materials from a single supplier is actively conducting single-source procurement. This procurement type is encouraged when the supplier and the client have a good relationship with each other. Single-source procurement enables faster procuring with expected quality because of only one supplier. It also offers lower pricing since all requirements are consolidated with one supplier (Mckinnon, 2023). It is also essential to use single-source procurement only if the supplier has past experience from similar projects (Oswego, 2025).

Multiple-source procurement

Multiple-source procurement refers to a construction company procuring materials and services it deems necessary from multiple different suppliers. It enhances business collaboration and quality because of the competition among suppliers (Overvest, 2025).

Competition between suppliers leads to lower prices, which is why companies often inquire prices from multiple sources to compare both pricing and quality before deciding whether to procure or not. Biggest advantage of procuring from multiple sources is the distribution of risk. By working with various contractors or suppliers, the company can mitigate potential risks and ensure a backup plan in case any participant in the supply chain encounters difficulties such as bankruptcy (Mckinnon, 2023).

Two-stage tender

In a two-stage tender, a construction company provides potential contractors or suppliers with a set of requirements, who then submit their pricing offers. After reading through the initial bids, the company asks the most promising candidates to submit a more detailed offer. This approach is often favored for complex procurements, as it allows the company to gain a clearer understanding of pricing and the subcontractor's understanding of the project before making a final decision (Mckinnon, 2023).

For clients, the benefit of what a two-stage tender offers are the involvement of contractors or suppliers early in the process. This minimizes risk and gives more certainty regarding the prowess of the contractor or supplier (Jorge & Lynch, 2024).

Open tendering

Open tendering is a process in which a construction company invites all interested suppliers to submit bids for a project. In this method, a refundable deposit from all applicants may be required. When conducting open tendering, the construction company is at risk of choosing the cheapest contractor but not getting the outcome they wanted in terms of quality (Autodesk, 2024b). This is because the contractor may be cheap, but their expertise might be deficient. To avoid this, past performance of possible contractors should be examined if possible.

Open tendering is typically used when the project's scope is clear and defined. A broad pool of suppliers capable of delivering the necessary materials and services need to be interested in order to conduct open tendering (Mckinnon, 2023).

Selective tendering

Selective tendering is a process where a construction company invites only pre-qualified and approved suppliers to submit bids for a project. This way, participation is limited only to reliable and skilled contractors (Autodesk, 2024b).

This method is typically used for demanding and sensitive projects where the company seeks to limit the number of suppliers with access to detailed information. Selective tendering is ideal for contracts that require specific expertise, as only a handful of contractors may meet the standards of the client. By reducing the amount of bidders early on, the quality of bids improves and helps to accelerate construction time and lower procurement costs (Mckinnon, 2023).

Negotiated tendering

Negotiated tendering is a process where a construction company works directly with a supplier to come on terms on a contract. This accelerates the procurement process because there is no need to go through the formal bidding phase (Mckinnon, 2023).

Negotiated tendering should be used in types of contracts where timeline of the project is urgent, the two parties have long history of working together, and if the contractor is involved in funding the construction project at hand (Autodesk, 2024b).

In conclusion, the most suitable procurement method for a construction project depends on its specific needs. Every client aims to choose a strategy that gives all the parties a suitable contract in terms of value and profitability (Mckinnon, 2023).

2.2 Procurement Process in Construction

Construction procurement can be defined in multiple ways, but all of these definitions are closely related in terms of meaning. Out of all these definitions, there is a consensus among field professionals that procurement process involves a series of activities and steps through which clients acquire specific services and materials related to construction and engineering works within a given time period, agreed terms and cost (Ibem & Laryea, 2014).

Before procuring for a construction project can begin, procurement packages for all work phases must be prepared. The creation of a procurement package requires a procurement plan which consists of a procurement list, schedule and assigned procurement responsibilities. The procurement plan is finalized immediately after the general schedule is completed at the start of a construction project. The creation of a procurement plan enables the formation of procurement packages which allows for structured and cost-effective procuring (Rakennustieto, 2010).

In construction, operational procurement can be linked to project-based procurement process. It consists of a construction company procuring all materials, equipment and services for an individual construction project based on previously composed procurement packages. All the processes in project procurement are specifically custom made for the project's requirements. The duration for project procurement is as long as the construction projects duration and its scope is limited on the specific needs of the project (Nouti, 2024).

Each phase of a procurement process consists of different responsibilities between multiple parties during the lifecycle of a construction project (Ruparathna & Hewage, 2015). According to (Ibem & Laryea, 2014), procurement process can be broken down into six basic activities as outlined in the international organization of standardization – ISO 10845-1. These activities

can be seen from Figure 3 below where the procurement process starts with the establishment of what is being procured and ends with the administration of contracts ensuring that the supplier complies with the set requirements.

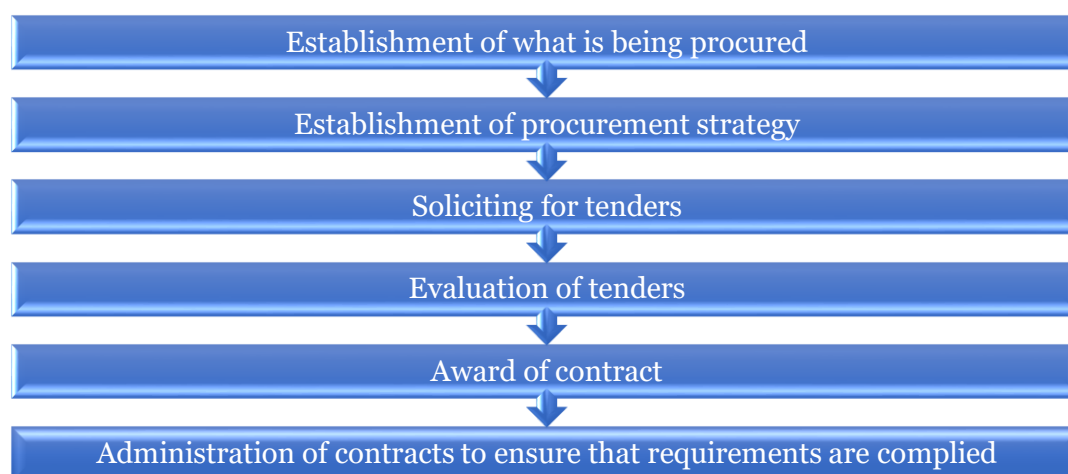


Figure 3. Six basic construction procurement activities according to ISO-10845 (Ibem & Laryea, 2014).

2.2.1 Establishment of what is being procured

Procurement in a company starts by choosing a procurement strategy for an individual project. This consists of identifying the objective, constraints and complexity of the project at hand. Specifications on what is being procured need to be clear before any action is taken since the success or failure of the project is based on choosing an appropriate procurement strategy (Sani Bako, 2016).

Assessing the need for what needs to be procured varies between direct procurements and more complex multi-stage procurements. In direct procurements, such as ordering material from a hardware store, the need is often clear. If the need is more complex, such as procuring a security service to the

construction site, the requirement specifications are more demanding (Logistiikanmaailma, 2023a).

In instances where procuring is not straightforward, it is the procuring personnel's responsibility to identify the possible challenges and abilities of the possible suppliers (Logistiikanmaailma, 2023a). Procurement teams work closely together with stakeholders to identify and gather detailed requirements to ensure that the upcoming project goes according to expectations. This can be technical specifications, quantities and quality standards (ProcureAbility, 2025).

2.2.2 Establishment of procurement strategy

Companies need to select one suitable procurement method from among the six most common methods explained earlier. Construction projects often require subcontractors with specific expertise, which is why selective tendering is commonly used in subcontracting procurements. By reducing the number of bidders early on, the quality of bids improves. This accelerates construction time and lowers the overall procurement costs. Still, this is not always so straightforward. It is up to the construction companies needs what kind of procurement methods they choose (Mckinnon, 2023).

Companies need to be aware of the current status of suppliers. Preferably, they should have a pre-existing network of competent suppliers to choose from. This makes it easier to analyse the supplier market and make responsible procurement decisions (Watts & Hahn, 1993). This is why bigger companies utilize shared databases to update supplier information, which can be leveraged for future use. These databases provide insight into whether the company has previously collaborated with a potential bidder or not and what the performance has been in the past (Logistiikanmaailma, 2024d).

Utilizing these databases makes it easier for the companies to be aware of the current status of the supplier market. Old suppliers may have downgraded in terms of quality and reliability while new ones may have been on a steady rise receiving recognition through impressive quality and low costs.

2.2.3 Soliciting for tenders

Request for Quotation (RFQ) is sent for the most promising suppliers that have successfully passed the previous stages of the project procurement process. RFQ asks the supplier to give the company an offer based on all the information provided such as blueprints, quantities and timelines (Logistiikanmaailma, 2024b). The request for quotation should be as precise as possible and it should explain in what kind of form the tender is expected to be made. The person sending the RFQ to the suppliers is ultimately responsible for the contents of the documents (Rakennustieto, 2010).

2.2.4 Evaluation of tenders

The received tenders are evaluated by conducting a bid comparison. Bid comparison consists of analysing bids after the tender submission deadline. Conducting bid comparison is important because it ensures that the company gets the best possible supplier concerning the specified job requirements (Autodesk, 2024a).

Generally, comparison of bids should be based on price, quality, and delivery time. This works only for the most basic procurements because criteria such as environmental requirements and sustainability have become increasingly important and more popular. This affects procurement decisions because the cheapest offer may not always meet the standards set for the initial procurement. The goal of bid comparison is to find out which supplier can carry out the project as stated in the RFQ with the timeline set and with good quality. Procurement personnel weigh all these factors together in various meetings and come to a conclusion which leads to negotiations with the most

promising supplier (Logistiikanmaailma, 2024c). It should be acknowledged that the lowest bid isn't always the most optimal option. It is advised to also consider the possible suppliers reliability, past performances and overall performance (Autodesk, 2024a).

2.2.5 Award of contract

If a supplier is procured for a specific project, it's time to move on to contract negotiations. Negotiating the terms of the contract is one of the most critical stages in successful procurement since it defines the legal framework for everything that follows in the implementation phase (ProcureAbility, 2025).

Contract negotiations ensure that both sides have a mutual understanding of the responsibilities and obligations included in the contract. Negotiations should be conducted with one supplier at a time, and another negotiation partner should only be selected if the negotiations with the first one do not result in an agreement (Rakennustieto, 2010).

Companies have a variety of different contracts such as project specific contract, yearly contract, and one-time contract. Project specific contract is made separately on every project and its contents and duration is determined on the negotiations (Logistiikanmaailma, 2024a). A legally binding and final contract will become active as soon as the supplier acknowledges and signs it (Blaney, 2025).

Contents of a contract between a company and the supplier include for example contracting parties, price, definitions, specifications, delivery time, validity period, warranty, and sanctions. General terms and conditions are always incorporated to the contract to safeguard the interests of the companies (Logistiikanmaailma, 2024a).

2.2.6 Administration of contracts

After the contract has been agreed upon and signed, the phase of overseeing the supplier relationship begins where the supplier starts to deliver its goods and services within previously agreed upon timeframes (Blaney, 2025). During this time the supplier is constantly evaluated. This consists of making sure that the quality is as agreed upon, work is on schedule and there are no unresolved problems (Logistiikanmaailma, 2023b).

If any problems arise during the relationship between the company and the supplier, it is the company's responsibility to notify the supplier as soon as defects are noticed (Blaney, 2025). This is why the company, and the supplier should agree on regular feedback sessions and reviews to assess the progress and overall performance, and if any precautionary measures should be taken (ProcureAbility, 2025).

2.3 Challenges in Procurement

Construction industry, among many other industries, has gone through substantial changes in the modern era. Materials, companies, and methods have all been able to meet ever evolving standards and legislations but at the same time the ability to obtain materials and services needed for construction projects keeps being more resistant to change. Construction procurement is a major contributor to trying to control construction projects' costs and at the same time ensuring profitability for contractors (Vecchio, 2024).

Between different stakeholders in a project, the construction procurement system faces multiple collaborative challenges located mainly at the knowledge management level (Ghazali et al., 2012). This is due to excessive information overload where the information received at one level may not always reach all the parties. Figure 4 illustrates how this happens by showing how suppliers, sub-contractors, contractors, site owners, engineering firms

and consulting organizations collaborate at the procurement level (Ghazali et al., 2012).

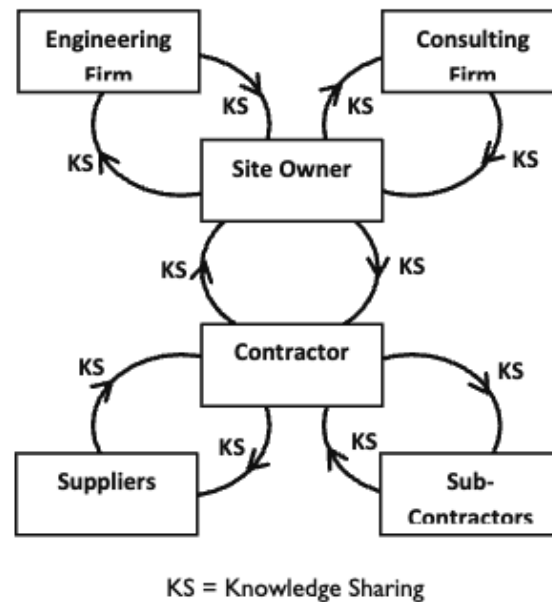


Figure 4. Collaboration and information sharing in procurement processes (Ghazali et al., 2012).

Construction differs from other industries due to its distinctive nature. Construction industry is becoming more complex with the advancement of technologies which involves several parties communicating with each other in a construction project. These advancements have highlighted the importance of effective communication. (Fahad & Mohamad, 2023).

Streamlined communication is essential in construction projects, especially during the procurement process, since it plays a big role in the success of a project. Clear information exchange between clients, subcontractors, and suppliers determines the success of bidding, contract negotiation, and delivery. Miscommunication can lead to project delays, legal issues, and failures. In addition to these, more severe ramifications can occur if information is not

circulated properly, such as litigation and arbitration (Fahad & Mohamad, 2023).

Because of procurements' nature and constantly changing circumstances in projects, challenges can be identified everywhere. Additional challenges presented by (Ghazali et al., 2012) at procurement level include:

1. Short-term commitments of project players
2. Lack of trust between the contractor and the supplier
3. Clashes and disagreements
4. Closed relationships and difficulties in sharing documentation
5. Uncontrolled additions and modifications
6. Unclear roles and expectations between parties
7. Cost and timetable overruns
8. Blaming others instead of focusing to resolve problems

2.4 Digitalization in Procurement Processes

Digital technologies are being used in an increasing manner in construction procurement processes. Digitalization has the potential to bring new business models and technologies to the disposal of companies, and this can be seen as new products and services. These technologies that have qualifications to enhance and grow companies business models are automation, robotics, machine learning, artificial intelligence, virtual reality, data analysis, and distributed ledger technology (TyEl-ministeriö, 2025). Despite the efforts in utilizing digital tools, there has been no systematic attempt to categorize different tools and technologies to carry out procurement digitally in the construction sector (Ibem & Laryea, 2014).

2.4.1 Digital Procurement

The concept of digital procurement has just emerged in the recent years. It is characterised as a concept that involves automation of procurement process by taking advantage of e-procurement systems that have been integrated with smart analytical tools like data mining solutions and big data (Ilhan & Rahim, 2020). E-procurement refers to the usage of e-commerce technologies to streamline a company's purchasing processes and it aims to enhance and automate as much as possible the procuring activities of the company (José Garrido-Samaniego et al., 2010). Generally, the majority of e-technologies are pre-existing web-based applications which have the ability to enable real-time communication and collaboration across construction supply chains (Ibem & Laryea, 2014).

The integration of Information and Communicating Technologies (ICTs) within construction processes has improved certain project collaboration factors during the last decade. These improvements can be seen both at the intra-organizational level and within a supply chain context. These technologies include for example electronic data interchange (EDI), supply chain optimization software (SCO), web-based integration systems, radio frequency identification (RFID) systems, and enterprise resource planning (ERP) systems (Ghazali et al., 2012).

E-procurement technologies and applications available to support the basic six procurement activities are in the form of web-enabled/supported software packages. BIM can be used for creating construction documents when establishing what is being procured. Web-based Enterprise Resource Planning (ERP) can be used when establishing a procurement strategy. When soliciting for tenders, web-based project portals, web sites, and cloud technology can be used. Evaluation of tenders could benefit from video conferences and cloud technology. Award of contracts can be operationalized through e-mail technology or wireless technology. Administration of contracts could be

enhanced by using e-marketplaces, electronic data exchange, and wireless technology (Laryea & Ibem, 2014).

In addition, The Finnish State Treasury has launched a project to standardize messages used in the order to delivery process by adopting PEPPOL-compliant messaging framework. PEPPOL (Pan-European Public Procurement Online) is a Europe wide platform that allows different organizations to share electronic business documents using a standardized format that works across all industries. According to preliminary estimates commissioned by the State of Treasury of Finland, buyers and suppliers in the construction value chain could save up to tens of millions of euros annually by switching to electronic message exchange (Peltokorpi et al., 2023).

The standardization and streamlining of digital procurement processes has been identified as a key strategic priority for organizations to improve efficiency in procurement processes. Since main contractors and subcontractors serve a wide range of clients across both public and private sectors, it is advised to establish a unified and simplified procurement system architecture. This could promote broader adoption of digital procurement tools and reduce repeated investment costs in separate systems (Yevu et al., 2021). This claim is further strengthened in a study conducted by (Mikkonen et al., 2019) where it is stated that transitioning away from fragmented and poorly integrated systems toward a distributed data-sharing model is essential. Manual data integration is prone to errors, results in slow information transfer between systems, and often leads to increased operational costs.

Many field experts say that e-procurement tools offer various types of advantages to the companies using them. These advantages can be operational, strategic or tactical but they can be only achieved through careful implementation. (Rahim, 2014). Digital procurement is crucial for the construction industry as it aligns the building schedule with inventory management and improves supply chain performance (Ojo- Fafore & Laryea, 2024).

At the operational level benefits can be seen related to procurement process efficiency and effectiveness with also including process cost savings on transactions. Strategic benefits show as enhanced reputations and improvements in customers supplier relations. Tactical benefits enable faster payments, improved control and monitoring, and increased efficiency (Rahim, 2014). (Leme, 2024) states that the benefits of e-procurement tools can be seen in companies through process standardization, containing costs, enhancing visibility and control, and accelerating integrated procurement.

2.4.2 Challenges in Digital Procurement

Despite the advantages that e-procurement tools offer to companies, the integration of them into procurement hasn't happened seamlessly and without challenges. From a technological aspect, many construction companies don't have suitable tools to process the data from procurements which leads to challenges in organizing information. The challenges faced by the construction industry in adopting new procurement innovations come from a misalignment between innovations and industry requirements, as well as the limited interest in adapting sector-developed ideas by software developers (Laryea & Ibem, 2014).

Although there is a clear need to connect procurement activities, stakeholders, and technologies, a wide-spread integration is still lacking. A key barrier is the inability to combine various web-based tools into a single system that supports the entire procurement process between all stakeholders (Laryea & Ibem, 2014). Currently there are multiple digitalized procurement tools in the market available for construction companies to use. Often, these tools lack compatibility with each other which means that they can't successfully execute all of the phases of procurement process, leading to combined usage of different tools. Other problem is related to insufficient interoperability between different systems which can be seen through difficulties on obtaining information from one application to another. (Ibem & Laryea, 2014).

According to (Ibem & Laryea, 2014), these challenges can be mitigated by encouraging digital technology developers to focus more on applications that enable procurement from a single platform, reducing the need for unnecessary switching between applications. Secondly, they state that the underlying cause of interoperability issues between different systems is related to varying cloud environments and e-procurement platforms. Future digital technologies should facilitate seamless data exchange and application migration across platforms through standardized cloud-marketplace systems. This would ensure compatibility across various vendors and ecosystems.

Conclusively, if the construction industry wants to enhance its productivity and efficiency it needs to start focusing on investing in digital tools and technologies that will offer the greatest operational impact and value. Additionally, there is a need to start providing training and education programs to develop necessary digital skills (Ojo- Fafore & Laryea, 2024).

3 Generative Artificial Intelligence and Construction Industry

3.1 Background

As stated before, artificial intelligence refers to a computer program that can perform operations deemed intelligent. It has different subsets that are interconnected with each other e.g. machine learning (ML), deep learning (DL) and generative artificial intelligence (GenAI) (Torro, 2024).

Machine learning is a subset of AI that learns based on experience without explicit programming. Deep learning is a more advanced subset of machine learning that processes vast amounts of data and imitates human brain with the help of neural networks (Torro, 2024). Generative artificial intelligence refers to a form of deep learning that can create original content. This content can be text, images, software code or audio created through a prompt provided by the user (Stryker & Scapicchio, 2024). Figure 5 below demonstrates how all these subsets are constructed (Torro, 2024).

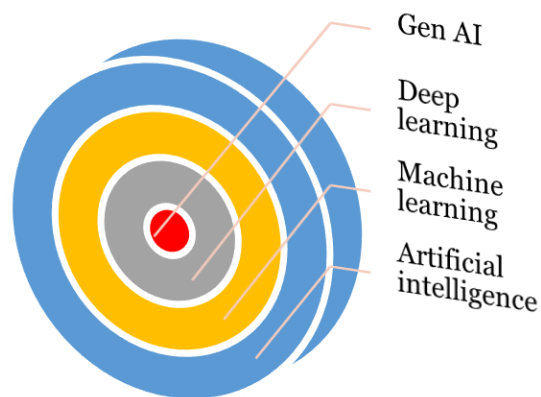


Figure 5. Different subsets of AI (Torro, 2024).

Since 2010, generative AI has had a breakthrough within the field of artificial intelligence (Torro, 2024). It is a branch of AI that is used to create novel and

realistic data or content, and it aims to produce data or content that is not only realistic but also expressive, diverse, and original (Taiwo et al., 2024). Generative AI uses large language models (LLMs) and multimodal GenAI models. Both of these are neural network models that are trained on extensive datasets and they have different capabilities (Taiwo et al., 2025):

1. Multimodal GenAI models are machine learning models trained on a wide spectrum of unlabelled and generalized data. These models are capable of performing a wide variety of general tasks such as image generation.
2. Large language models, such as OpenAI's ChatGPT, are one subset of foundation models. LLMs are specifically used to summarize and generate text and extract information.

These models are engineered to simulate the learning and decision making of the human brain. The deep learning models learn from the data uploaded to them which makes them able to produce logical and well-organized content. Ultimately, this content feels like it was created by a human (Stryker & Scapicchio, 2024).

3.2 Generative AI operating principle

Generative AI operates in three mutually dependent distinctive phases as can be seen from Figure 6 below.



Figure 6. Generative AI's operating phases (Stryker & Scapicchio, 2024).

Training is the first phase which consists of creating a foundation model that serves as the base foundation for multiple gen AI applications. After this, the foundation is tuned to a specific GenAI application. Lastly, the model is used, evaluated and returned to assess its input and continued to improve its accuracy and quality (Stryker & Scapicchio, 2024).

3.3 Benefits of Generative AI in Construction

The most obvious benefit of generative AI is enhanced productivity. Because it can provide content and answers to everyone who demands it, generative AI has the fortunate ability to free employees' time for more time-sensitive tasks. Additionally, generative AI can potentially cut costs within the companies using it (Stryker & Scapicchio, 2024). For companies and business managers to be able to truly utilize GenAI to its full extent, mastering of prompt engineering is required (Robertson et al., 2024).

A study conducted by (Saka et al., 2024) shows how OpenAI's Generative Pre-trained (GPT) models, which are one example of a GenAI tool used in ChatGPT, offer opportunities for construction industry across phases as pre-design, design, construction, operation and maintenance, and demolition. Benefits can be found also in value-added services.

The potential benefits of GenAI based solutions are further strengthened by their emerging use cases in the construction industry. According to (Abioye et al., 2021) and (Nygqvist et al., 2024b) application areas and future opportunities of GenAI in the construction industry are estimation and scheduling, construction site analytics, voice user interfaces, risk assessments, and secretary tasks.

Table 1 below summarizes the key use cases and benefits of GPT models in different phases of a construction project while as Table 2 summarizes future application areas where GenAI can potentially be used.

Table 1. GPT model applications across construction project phases (Saka et al., 2024).

Project Phase	Use Case	Description
Pre-design	Procurement decision support	GPT models can enhance procurement decisions by evaluating data and providing suggestions that steer procurement personnel to more accurate, data-driven decisions. In addition, textual analysis capabilities can be used to gain insights on supplier reviews. Also, BIM can be integrated with a GPT-model and the procurement database, enabling queries related to procurement and automating workflow such as status updates and order placements (Saka et al., 2024).
Design	Quantity take-off and costing	GPT models can automate quantity take-off by providing information about a project's materials, design, and other specifications. GPT can prepare bills of quantities and analyse past cost data for predictions. Also, it can be linked with Python to assist in extracting quantities from BIM models (Saka et al., 2024).
Construction	Project schedule and logistics management	GPT models can be effectively used to acquire and analyse textual information which makes it a notable tool to be used in construction project scheduling and logistics since it offers a more comprehensive understanding of their complex nature in a construction project (Saka et al., 2024).
Operation and maintenance	Regulatory compliance management	GPT models can be trained to capture patterns and embedded knowledge with compliance documents, guidelines, and reports. These models can be used to automate compliance checks, identify non-compliance issues, and create reports. This saves time and ensures constancy within a project (Saka et al., 2024).
Demolition	Materials recovery planning and maximization	GPT can add to this phase through its ability to optimize recovery and recycling paths and by supporting high-value material allocation. This will rise probability of resales and proper disposal of hazardous materials (Saka et al., 2024).
Value-added services	Stakeholder communication	Stakeholders of a construction project need to collaborate and share information daily to ensure a smooth delivery of a project. GPT models can personalize reports and project updates for all the stakeholders daily to keep up with project advancements. It can be also used to summarize long documents to gain an understanding of a given topic which are discussed in meetings (Saka et al., 2024).

Table 2. Future opportunities for GenAI in construction (Abioye et al., 2021; Nyqvist et al., 2024b).

Application area	Opportunities
Estimation and scheduling	Use of deep learning models for cost and time estimations, planning of work schedules (Abioye et al., 2021)
Construction site analytics	AI chatbots for information flow, AI-driven analytics for quality control (Abioye et al., 2021).
Voice user interfaces	AI-driven voice user interfaces for on-site and off-site activities (Abioye et al., 2021).
Risk management	Risk identification, risk analysis, cost overrun analysis, and definition of risk management procedures (Nyqvist et al., 2024b)
Secretary tasks	Transcription of meeting recordings and comparative analysis of documents and past meetings (Nyqvist et al., 2024b).

GenAI is without a doubt a notable tool in the evolution of risk management practices to enhance human performance (Nyqvist et al., 2024a). By applying it to the construction field offers good potential in transforming its conservative practices by streamlining project management tasks, mitigating risks, and enhancing decision making processes. It must be noted that to do so, the workforce must be comfortable with the concept of using GenAI with everyday tasks. This can be achieved by investing in skills development through various training seminars and continuous learning. (Nyqvist et al., 2024b).

3.4 Challenges of AI and GenAI in Construction

Integration of AI into construction questions the old ways of working and may cause resistance for change (Torro, 2024). There are cultural and logistical barriers that make it hard to integrate solutions like AI to its processes. This can be explained with how people have a hard time to adapt to continuous developments in technology which leads to wasted potential. These

challenges could be mitigated by improving education and support regarding AI integration to company processes (Peltokorpi et al., 2025). AI's human like abilities make it an entirely new technology with limited prior experience. This further demonstrates why its effects on workers attitudes on long term cannot be projected (Torro, 2024).

AI technologies to be integrated into construction industry need to be tested and used in various projects before insisting on all stakeholders on using them (Abioye et al., 2021). This creates trust and transparency between all parties and makes the integration process easier. Stakeholders are more prone to adopt AI technologies to their usage if it has been successfully used and tested by others. This kind of scenario is problematic to achieve since construction industry is currently facing a significant shortage of professionals with prominent AI usage skills. This makes it challenging to develop and implement AI based solutions to specific needs (Peltokorpi et al., 2025).

Artificial intelligence's reliability is also something that needs to be reviewed internally within companies since even the creators of AI models don't know how their models come up with certain answers (Torro, 2024). This is called a black-box approach which means that even most of the AI models don't explain the reasons for reaching conclusions (Abioye et al., 2021). The black box phenomenon further amplifies the resistance to new AI solutions since it lacks transparency and hinders trust and acceptance among construction professionals (Peltokorpi et al., 2025).

GenAI based tools suffer from poor quality, fake content, and inaccurate information which may lead to making poor decisions on the worksite (Peltokorpi et al., 2025). This becomes a problem when individual workers start to rely on AI too much. This may lead to blindly trusting content provided by AI without checking its contents and reliability from other sources. The declination for the need to use own output in different tasks is not what AI should be used for (Torro, 2024).

As GenAI keeps on taking a more prominent role in decision making processes in the construction field it is important to focus on ethical considerations and the establishment of effective governance frameworks. Responsibility, transparency, and fairness are all notable concerns when implementing AI-based solutions to industries (Peltokorpi et al., 2025).

It is also important to acknowledge the regulatory and standardisation issues related to the integration of AI in construction companies. Currently there is a lack of consistent regulations to guide AI users to a responsible way of using it and the lack of standards for digital technologies in construction further adds complexity and problems to its integration. This is a notable challenge because at the moment professionals in the field of AI in construction companies struggle to understand and predict the features and limitations of AI. Misunderstanding these factors may lead to improper usage of AI-based tools and to errors and decreased performance (Peltokorpi et al., 2025). Table 3 next page summarizes the above key challenges identified in recent studies along with their description and sources.

Table 3. Challenges of AI and GenAI in construction

Challenge	Description
Resistance to change	AI challenges traditional ways of working and may face cultural/logistical resistance (Torro, 2024).
Lack of education and support	Insufficient training leads to underuse and misunderstanding of AI tools (Peltokorpi et al., 2025).
Uncertainty of long-term effects	AI's human-like nature creates unpredictable impacts on attitudes (Torro, 2024).
Lack of testing and trust	Stakeholders are reluctant to adopt unproven technologies (Abioye et al., 2021).
Shortage of AI-skilled professionals	Limited expertise slows down development and implementation (Peltokorpi et al., 2025).
Black-box phenomenon	This phenomenon lacks transparency and hinders trust and acceptance among construction professionals (Abioye et al., 2021; Peltokorpi et al., 2025).
Inaccuracy and misinformation	GenAI may generate poor quality or false content which affects decisions (Peltokorpi et al., 2025).
Overreliance on AI	Workers might overly depend on AI, reducing critical thinking (Torro, 2024).
Ethical concerns	Lack of fairness, transparency, and responsibility in AI systems (Peltokorpi et al., 2025).
Regulatory and standardization issues	Inconsistent regulations and missing standards complicate usage (Peltokorpi et al., 2025).

GenAI has the potential to change everyone's perception of productive working but to fully realize its benefits, it is essential that it is developed in safe and responsible environments (Peltokorpi et al., 2025). Before all, AI should be a helpful tool to streamline processes, but it should not delete peoples own thinking and decision making. Achieving this is a challenge but the potential of integrating AI-based solutions to industry processes is positive and promising.

3.5 Integrating Generative Artificial Intelligence into Business

Artificial intelligence and humans have a reciprocal relationship. GenAI needs people's prompts and offered data to evolve, and humans need it for streamlining processes. For this to work, a very user based approach with

company's workers is required when innovating new AI-based solutions (Torro, 2024).

Artificial intelligence offers companies value through new or altered business processes and increased productivity. This means streamlining processes by reconstructing them completely or enhancing only the wanted parts. In an instance where AI is used only to enhance productivity by incorporating AI based customer service chatbots to company websites, processes' reconstruction is not needed (Torro, 2024).

If a company wishes to succeed in integrating AI into its business, it needs to be technologically mature and have interest in standardizing AI into its processes (Torro, 2024). This requires open dialogue between the management and workers. Everyone one in the company should know what AI is, what it is projected to be used for and that it is not here to take away jobs but to rather offer a toolbox to enhance ways of working.

According to (Andrushchak & Kondakov, 2024), the integration of GenAI into company processes helps them to enhance their customer experience, business continuity, revenue growth, and cost optimization. Customer experience is the leading reason why companies want to adopt GenAI to their businesses with a share of 38%. Next is revenue growth with 26%, cost optimization with 17% and business continuity with 7%. 12% of respondents felt that none of these alternatives should not be enhanced with GenAI. This is illustrated with a percentage chart which can be seen from Figure 7 below.

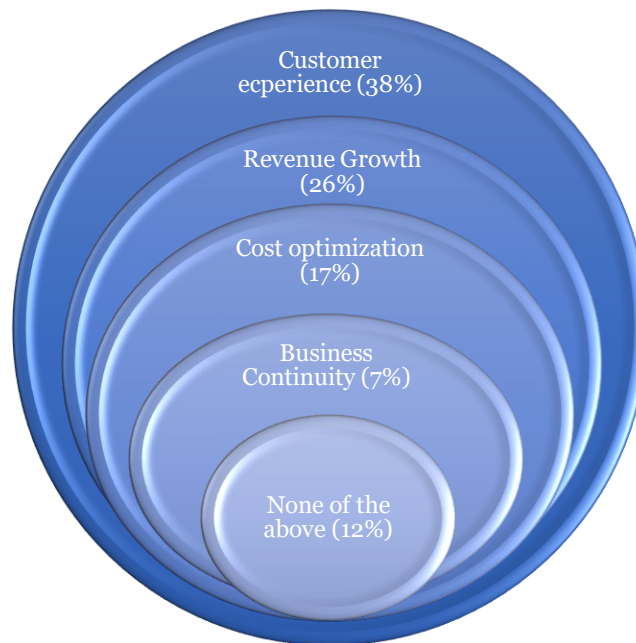


Figure 7. Why companies want to adopt GenAI to business (Andrushchak & Kondakov, 2024).

Optimally, AI will discard nonvalue adding tasks and overlapping processes which leads to new more efficient ways of working. Unfortunately, with all the precautions and careful planning, many attempts to incorporate AI into pre-existing processes in companies fail due to non-technical factors such as poorly defined scope (Torro, 2024).

3.6 Prompt Engineering for Gen AI

Prompt can be defined as the information, questions, or sentences entered to a GenAI tool that in return provides the user with responses based on the given prompt (Harvad University, 2023).

An engineered prompt is as good as the preparation done in engineering it. Prompt engineering entails thoroughly thinking through the process of how to construct the problem, opportunity or question that will be input to the GenAI tool (Robertson et al., 2024). A well-structured prompt can

significantly improve the precision, relevance, and quality of outputs provided to the user (Cate, 2025).

3.6.1 Engineering of a Prompt Through Co-Construction

By establishing a collaborative relationship between the user and GenAI, a well-structured prompt facilitates knowledge co-construction and enhances information retrieval. Co-construction can be explained through constructivism where knowledge is obtained through an iterative process where past experiences shape future research (Robertson et al., 2024). Figure 8 illustrates this learning cycle where a prompt is generated through various steps only to end up back in the beginning if the result is not as precise as wanted.

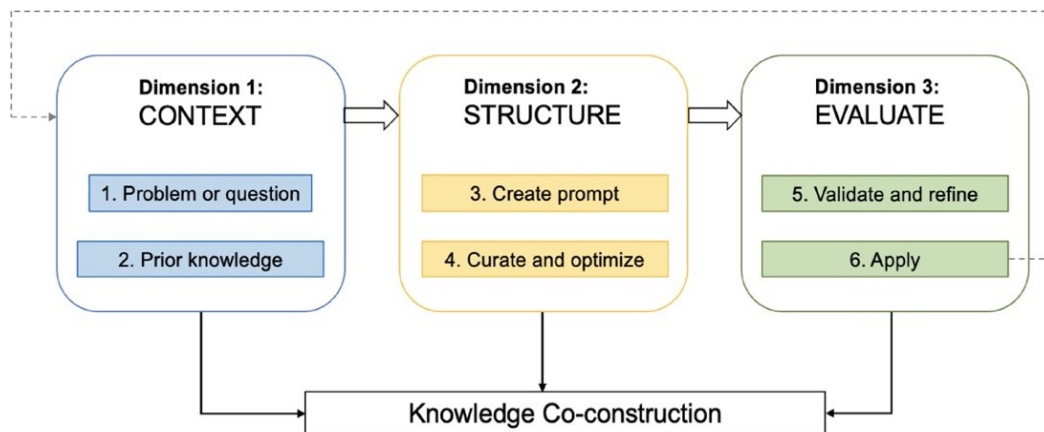


Figure 8. Creating a prompt through knowledge co-construction (Robertson et al., 2024).

Prompt generation starts with acknowledgment of the context. Context provides the GenAI model background information to help it understand its task and to ensure that it generates relevant and targeted outputs (Cate, 2025). Current problem, question, and prior knowledge are the foundation for generating a prompt that offers an answer with value.

In the structure phase, the model needs a clearly specified instructions on how it should organize and present its answer (Robertson et al., 2024). This can be the tone, context, scope of the response, example or style (Cate, 2025).

Evaluation assesses the quality of the models answer. When evaluating the response provided by the GenAI, users need to evaluate the answers quality but also their own bias in the application of the output (Robertson et al., 2024). If adjustments are needed, re-evaluation of the context and structure is required, thus highlighting the importance of knowledge co-construction.

Prompt generation is an essential skill to practice as often as possible if one is trying to achieve its full benefits. With a good prompt, there is a better chance that the full capability of the used AI model is unlocked. This leads to better results from the AI, which in return enhances productivity and creates possibilities for solutions not before thought possible.

3.7 Conclusions From the Existing Literature

The literature review addressed the research questions set for this thesis. The benefits of digitalization tools and the challenges and biases regarding their usage was discussed. GenAI's potential applications in procurement processes were also identified.

The literature review showcases how construction industry is a conservative field that always seems to be one step behind in terms of utilizing new technologies. Still, digitalization has the potential to bring new business models and solutions to the disposal of companies, such as AI based solutions. At operational level the benefits of digitalization can be seen related to procurement process efficiency and effectiveness with also including process cost savings on transactions. Strategic benefits come across as enhanced reputations and improvements in customer supplier relations. At the tactical level digitalization can be seen as improved control and increased efficiency.

The potential challenges and biases regarding AI integration into procurement workflows can be explained by construction companies not having suitable tools to process procurement data which leads to challenges in organizing information. There is also a lack of training and education programs which further confirms why knowledge of such tools is not identified among field professionals. In addition to these, challenges faced by construction industry in adopting new solutions come from a misalignment between innovations and limited interest towards sector-developed ideas.

To tackle these challenges, it is essential for the construction industry to start focusing on investing in digital based solutions and technologies in order to obtain the greatest operational impact and value as possible. Also, organized training and education programs are called for to mitigate the biases regarding digitalization.

GenAI and its future usage in the construction sector indicates possible challenges due to its lack of originality and ability to produce human-like content. Still, its potential usability to perform risk assessment tasks and analytics similar to tasks performed in bid comparison was recognized. Also, the ability of GPT models to be integrated into BIM models is a substantial opportunity to enhance procurement package formation which is an essential part of the procurement process.

4 Empirical Research

The empirical research section for this thesis was carried out in two parts. First, interviews were conducted with the goal of identifying opportunities for utilizing AI in different phases of the procurement process at Skanska Talonrakennus Oy, and to identify potential challenges and biases regarding its utilization. Second, insights into how GenAI could be utilized to improve the accuracy and efficiency of bid comparison in procurement were obtained. This was carried out in a form of a case-study where a GenAI based bid comparison was compared to a manually performed one.

4.1 Data Collection and Analysis

This section first introduces the background, methods, and goal of the interview process. Then, the same is done for the case-study section.

For the interview study, five different procurement professionals from Skanska Talonrakennus Oy were interviewed face to face with each interview lasting for 60-90 minutes. The professionals varied in experience, age and position within the company which guaranteed a variety of answers from different perspectives and standings. Because of the variance in experience and job roles of the interviewed professionals, the responses from the interviews provided valuable insights into the current state and potential opportunities in procurement. The diverse perspectives and approaches to the questions contributed to a more comprehensive understanding of the subject.

The questionnaire was split into two parts, and it followed a semi-structured layout meaning that the questions were pre-determined, but the flow of the interview was more relaxed. Notes and transcripts were collected for analysis purposes. Goal of the interviews was to obtain an understanding of the current state of procurement processes and the potential of AI usage in it, and how GenAI could be used to enhance current way of doing bid comparison at

Skanska Talonrakennus Oy. A summary of the interview questions can be found in Appendix 1.

The interview data was analysed thematically by identifying and categorizing recurring themes for AI potential and challenges in different procurement processes. The findings were cross compared with the goal of finding similar and different views to the interview questions between different interviewees. After this, the answers were compiled to an understandable form and analysed. All of the findings were taken into account and used in the part 2 of the case study when generating the GenAI prompt.

The case study phase of this research consisted of developing a customized prompt for Skanska's own GenAI tool. The prompt was developed to be based on the conducted interviews, Skanska Talonrakennus Oy's guidelines, and scientific literature. Because this research was conducted for Skanska Talonrakennus Oy and their needs, the engineered prompt will not be thoroughly showcased in this research.

The GenAI system was tested with Skanska's computers and network with bids from 6 different subcontractors. These bids were all for the same task making them cross-comparable. The testing phase consisted of various testing sessions from where observations were made. In addition, feedback sessions with mentors from Skanska were held to steer the research in the right direction. The information obtained from the testing sessions was compiled to Skanska's own GenAI tool where it was at disposal when it was needed.

The prompt used for the testing emphasized accuracy, lookout for changes, and systematic approach based on pre-downloaded files. After testing the performance of the GenAI tool, the results were analysed and compared to a traditionally performed comparison with the goal of finding how GenAI can be utilized to improve the accuracy and efficiency of bid comparison in procurement. The comparison highlighted the accuracy, efficiency, reliability,

and potential of the provided outputs for integration into bid comparison. Accuracy and reliability were measured by seeing how close the GenAI tool was able to get compared to the manually performed one. Efficiency was evaluated by weighing the time it takes to perform a comparison to what kind of answer the tool provided. The results of this case study will provide insight into the potential integration of GenAI assisted bid comparison at Skanska Talonrakennus Oy.

It is important to highlight Skanska's own guidelines on how a professional and efficient prompt is engineered. According to (Skanska, 2025), a good prompt is as good as the wanted result and it consists of clearness, precision and relevancy to the topic. To ensure that a generative AI produces a reliable response, the prompt must be formulated clearly, provide precise details, and remain relevant to the topic. Skanska's own general guidelines for prompt engineering can be seen below. Figure 9 illustrates a good process to follow when creating a prompt for GenAI while as Figure 10 showcases an example of a good prompt following these process steps.

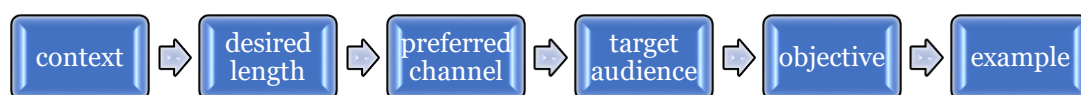


Figure 9. The process for creating a prompt (Skanska, 2025).

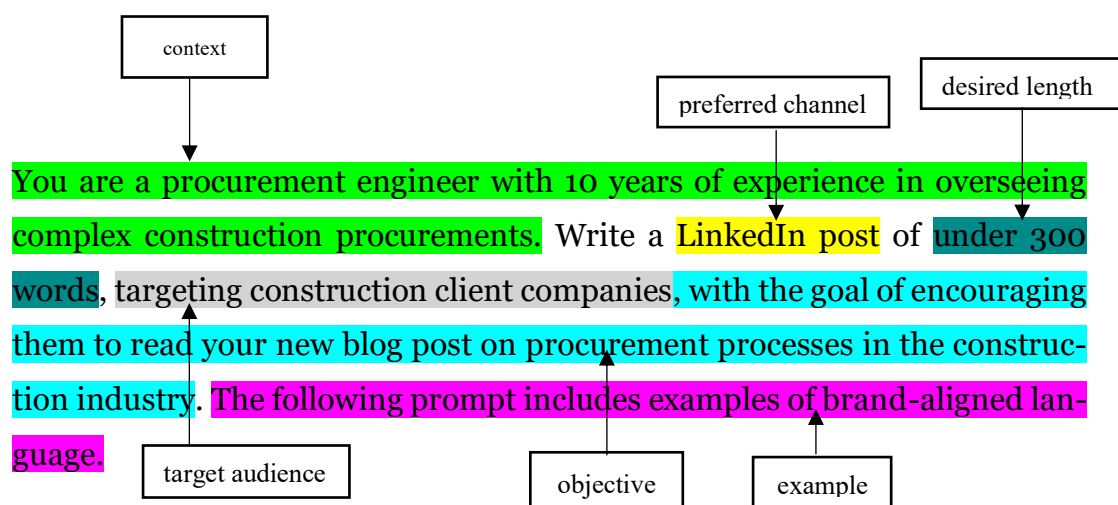


Figure 10. An example of a good prompt (Skanska, 2025).

The above example is a good starting point when generating a prompt for a specific task, but it should not be followed blindly. It is important to try different approaches when creating a prompt for a generative AI because it's the only way to see what kind of prompts work for an individual's specific needs.

4.2 Analysis and Results

4.2.1 Procurement at Skanska Talonrakennus Oy

To further demonstrate Skanska's way of doing procurement and justify the interviews, it is essential to showcase its process on conducting project-specific procurement. According to (Skanska, 2024), the key element of operational procurement is the project-specific procurement plan and accordingly conducted tenders. A project-based procurement encompasses all stages from procurement preparation to financial finalization. Figure 11 illustrates the main structure for the stages of project-specific procurement and where bid comparison lands in that process.

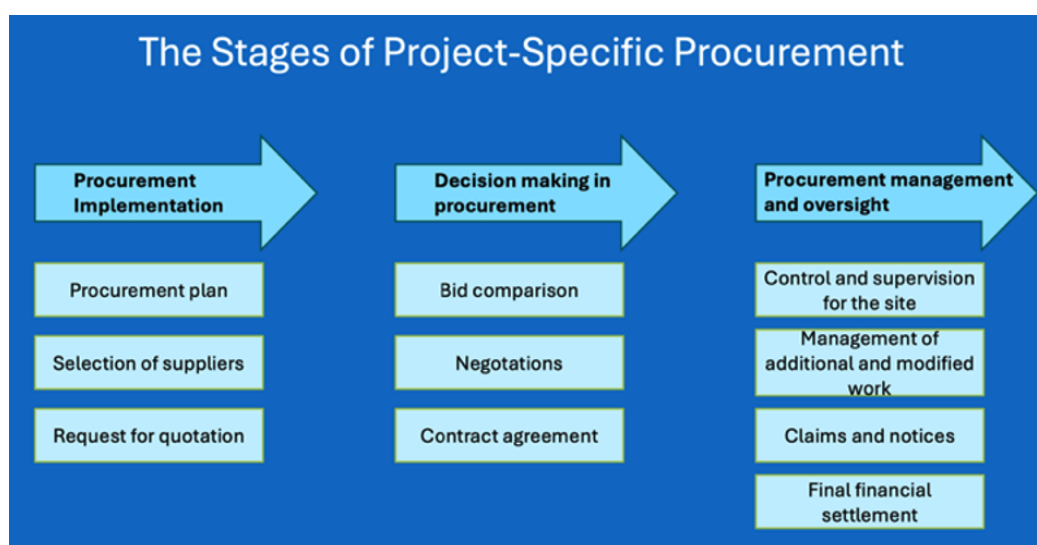


Figure 11. The stages of project-specific procurement at Skanska (Skanska, 2024).

As can be seen from figure 11 above, project-specific procurement at Skanska Talonrakennus Oy is a structured process that follows a standardized layout,

which every new Skanska employee is introduced to. It is also available for all employees on the company intranet. Despite this, all interviewed professionals identified manual phases that make the process time-consuming and challenging. These phases are presented in Table 4 below with the most notable phases mentioned being the generation of procurement packages and manual bid comparison.

Table 4. Identified inefficient manual phases in procurement.

Manual phases	Why is the phase time-consuming and challenging?
Generation of procurement packages	All documents for each tender are compiled manually
Bid comparison	Comparison is done manually in Microsoft Excel which takes time
Design gap identification	Interpreting information manually from multiple blueprints for the tender takes time
Bid layout analysis	Each bid is in different layout, and their analysis and interpretation separately takes time
Quantity calculation	Currently, quantity calculation is done manually, which takes time
Supplier selection	There is no certainty that old suppliers are active anymore which requires manual inquiries about their current status. This takes unnecessary amount of time from doing other tasks deemed more important.

Currently, procurement engineers are responsible for manually identifying project requirements, calculating quantities, and selecting potential manufacturers before procuring for a project can proceed any further. Two interviewees stated that it is common that in the early stages of a project the project designs are incomplete. Incomplete project designs force procurement teams to interpret missing information, increasing workload and risk for errors. These challenges highlighted discussions on the potential role of AI-assisted tools in procurement processes.

Several interviewees described Skanska's procurement process as systematic and effective. However, they observed a lack of standardization in bid

formatting and supplier selection criteria. As a result, it is common for subcontractors to submit bids in varying formats, even when they have received identical RFQs. This creates extra work in trying to get a comprehensive understanding of all the received bids and compare them equally. This is why potential standardization suggestions to the procurement process were discussed with the main goal of making the procurement process more productive. The suggestions were a standardized bid request template, clearer project specifications, and consistent contract structures.

The manual state of bid comparison was identified as a challenge in the interviews since it requires procurement teams to manually adjust and format bids for a fair evaluation. This challenge is further amplified when the received bids are all formatted differently.

Setting clearer project specifications was seen as a solution to avoid unclarity between supplier offers and standardized contract structures were suggested to prevent misalignment between supplier expectations and Skanska's common procurement terms.

It was stated that supplier availability is inconsistent with subcontractors sometimes declining bids due to high demand or the state of the market. Environmental and regulatory requirements were identified also as contract complicators. In the modern construction field these requirements are mandatory, but they still require additional time for making sure that the received bids follow Skanska's guidelines.

The interviewees emphasized how procurement often requires extensive communication with clients and project teams to clarify specifications for what it is that is being procured. Often, the communication between different stakeholders is scarce which leads to unclear scopes of what it is that needs to be procured. This slows down the decision making and makes the procurement processes needlessly slow. According to the professionals, this could be

solved by establishing and emphasizing clear communication between all parties in the earliest phase of the project as possible.

4.2.2 AI's Role and Opportunities in the Procurement Process

The interviews revealed that every interviewee saw strong potential for AI usage in procurement. Only difference between the professionals' views were on how AI should be integrated and what tasks it could handle. The professionals had previously encountered AI in supplier categorization and in simple tasks translating foreign-language bids. Based on the interviews, none of the professionals had encountered any categorized usage of AI in procurement processes.

The professionals' answers resulted in four main procurement phases where AI could provide value in. These can be seen from Table 5 below.

Table 5. Main procurement phases AI could provide value in.

Where could AI provide value in?	How to utilize AI in each phase?
Procurement package formation	Use AI-based quantity take-off tools to generate accurate material and cost estimates from BIM data.
Bid comparison	Automate bid comparison by extracting key data from the offers and end up with a comparison.
Contract oversight	Monitor contract compliance and milestones with AI-driven analysis of documents and schedules.
Supplier evaluation	Use AI to analyse supplier performance data and reviews to support data driven selection.

AI has the potential to automatize the bid comparison phase of the procurement process and at the same time save a lot of time and give opportunities for procurement engineers to do other work tasks. This would improve productivity of the whole process. In addition to this, AI could flag inconsistencies and highlight anomalies in the received bids' pricing structures. This way procurement personnel would make more informed decisions.

Procurement professionals highlighted manual workload as one of the biggest challenges in procurement. Currently they may spend an accountable amount of time researching suppliers and verifying their reliability. With AI-based supplier evaluation, there is potential that AI could assess suppliers past performance, compliance history, and contract fulfilment rates to aid selection. It could also check status for new suppliers and give an overall analysis of those.

The interviews showed that there are challenges in forming comprehensive bid requests as there are also problems in the layouts of the received bids. Currently all procurement packages are manually compiled and sometimes some details are missed leading to prolonged negotiations. For procurement package formation, AI has the potential to analyse project documents and generate procurement packages. This would reduce manual effort and give procurement professionals time to conduct other tasks deemed more important. One example of this is when one interviewee contemplated the potential usage of AI to analyse BIM-models. This way the AI could automatically organize material quantities from the model to an understandable listing.

The professionals stated that reviewing contracts is another time-consuming phase but still a needed one, since verifying contracts is essential to ensure compliance with regulations and company policies. Currently, contract terms are often reviewed manually which increases the risk of missing important compliance requirements. According to the interviewees, AI has the potential

to generate general contracts, assist in scanning contracts for missing clauses and overall, it has the potential to automate document handling and validate contracts.

One of the interviewees who is currently working in a manager role highlighted how AI could be used to enhance information sharing across Skanska's regional and business units. This would make procurement insights more accessible, enhancing productivity. For this to work, it was stated that AI generated insights' accuracy rate is critical since AI's effectiveness is only as strong as the data it is trained on.

Conclusively, the interviews showed different views on how AI could be used effectively in procurement processes. Some professionals saw AI as a tool that would reduce human workloads and improve decision makings accuracy. Other interviewee suggested that AI has the potential to detect hidden information from received bids.

4.2.3 Biases and Challenges in Using AI in Procurement

Even though the professionals identified areas for improvement and potential AI usage at Skanska Talonrakennus Oy, they also highlighted that it is important to acknowledge the possible biases and challenges and how to relate to them. This is important in a possible scenario where AI based solutions are actively being integrated to company processes.

The interviews revealed concerns that AI may not understand industry-specific regulations or contract details, which may lead to incorrect assumptions. These contract details were stated as environmental compliance regulations, supplier specific contract terms, and complex pricing structures that include variable costs. It was stated that months of development work and piloting is required before AI can be trusted to conduct procurement tasks on its own. Before wide introduction a piloting phase is necessary in one or two

construction sites. This would act as a test and proof for others that this kind of solution works. This would mitigate biases and make it possible for AI solution to take footing in the construction industry.

Some professionals were divided on how accurate and reliable AI would be in conducting tasks by itself. There was also a concern relating to AI's usage since it could lead to over trusting the AI tool which could lead to a loss of human expertise. This is why its usage should be prioritised as being a supportive tool rather than an absolute way of conducting procurement.

All of the interviewees discussed about the fact that construction industry is characterised as conservative and resistant to digitalization. This bias could be tackled by systematic introduction to possible AI-based solutions. This could mean shared introduction seminars, training hours, step by step integrations, AI mentors, and thorough instructions on how to use AI-based solutions. The most important aspect of integrating new solutions to company personnel would be not to make the user overwhelmed by information but rather fascinated by the opportunities it offers.

To measure AI's effectiveness in construction procurement, professionals identified five different performance indicators in the interviews which can help companies assess if AI-based tools are worth integrating to their processes.

The indicators were:

- Time savings and value addition
 - AI should be able to reduce manual workloads and accelerate procurement processes. Its success can be measured by whether procurement teams feel they have more time for strategic planning and value adding tasks rather than for repetitive work.

- Improved bid comparability
 - AI needs to correctly identify differences between bids and present clear comparisons.
- Decision making support capabilities
 - If an AI-based tool provides structured report that allow procurement teams to make informed decisions, it is doing what it is expected from it.
- Usability
 - Procurement professionals should start perceiving AI as an asset rather than an additional burden.

If AI integration and usage in company processes potentially has the ability to achieve time savings, the company will undoubtedly achieve enhanced productivity which ultimately shows as cost savings. This conclusion emphasizes how AI-based solutions are in construction companies' best interest.

4.2.4 GenAI in Bid Comparison

During the interviews, bid comparison was regarded as one of the most demanding and time-consuming tasks in procurement. It requires procurement personnel to manually format, analyse, and compare supplier offers with the goal of finding the most cost-effective supplier.

The interviewees stated that at Skanska Talonrakennus Oy, bid comparison is currently conducted on a predefined Microsoft Excel layout where procurement engineers manually format supplier offers, adjust pricing structures, and align bids with project needs. For a project-specific procurement there is a pre-reserved max allocation of funds that should be spent on a single procurement. For illustration purposes, the layout of an Excel based bid comparison can be seen from Figure 12 below.

All of the 5 interviewees mutually agreed that the current way of conducting bid comparison works and produces results, but it takes a great amount of time to do and there is still a potential for miscalculations. This is why AI-based solutions were a welcomed discussion topic.

The screenshot shows a Microsoft Excel spreadsheet used for bid comparisons. The layout is divided into several sections:

- Header Section:** Contains fields for 'Procurement:', 'Comparison creator:', 'Project:', 'Project number:', and 'Number of bids:'. There are also checkboxes for 'Request for Quotation (RFQ)' and 'Updated:'.
- Subcontractor Bids Table:** A table with columns for 'Subcontractor Name', 'A', 'B', and 'C'. Each of these columns has sub-columns for 'Unit Price' and 'Total Price'. The table has multiple rows for different subcontractors.
- Summary Section:** Located at the bottom, it includes a 'Sum:' row with values for 'Unit Price' and 'Total Price' for each subcontractor. Below this are rows for 'Additional and modification works (%)' and 'Additional and modification works:', followed by a 'Total:' row.

Figure 12. An example of the layout used for bid comparisons.

The interview sessions revealed that received bids from different subcontractors give a lot of work for the procurement engineers since they have to first analyse and comprehend them before manually setting them to a comparable form in Microsoft Excel. One example of this is when two subcontractors offer the same service but one of them has separately priced every minor detail while as the other has combined services and gives price based on those. This makes it hard for the procurement engineers to distinguish which one of the offers really is cheaper. This is further confirmed when the interviewees stated that suppliers often include additional costs to their offers, making direct comparisons difficult.

According to the conducted interviews, GenAI has the potential to improve bid comparison by most importantly conducting it, and by enabling automated bid formatting, and error detection. With bid formatting, GenAI can

convert all supplier bids into a standardized format and flag any missing bid quantities, hidden costs, or deviations from set RFQ terms.

GenAI also enables faster decision making where it can analyse large amounts of data in a fraction of the time a human could. This was seen as a first-hand opportunity if the GenAI tool could summarize all the offers to a short analysis package if the automated bid comparison would not work at the beginning of its integration. The AI-assisted analysis would enlighten the status for the procurement professional of what to expect from the bid comparison and help them to gain a basic understanding what the suppliers really are offering and at what cost.

Based on the interviews, GenAI has the potential to enhance bid comparison by automatising all of these steps and ensuring that all bids meet predefined standards before even starting its review.

Four out of five interviewees expressed and agreed upon their concern on that AI should not be fully automated in bid comparison in the early stage of its integration but rather serve as a decision support tool. All of the comparisons conducted by an GenAI tool should be reviewed by procurement professionals to ensure accuracy.

4.2.5 Engineering the Prompt – Case Study

The thought process for the prompt generation was to use Skanska's own GenAI tool to perform a comprehensive bid comparison from pre-attached supporting files. These files were Request for Quotation (RFQ) document, and all bid PDFs. These files laid the groundwork for the GenAI tool to recognize what needs to be procured, what are the offers, and what is the needed presentation layout for the comparison.

The following paragraph was used at the start of the engineered prompt. It showcases the prompts initial layout and purpose.

“Please generate a bid comparison from the provided PDF-file offers from our subcontractors. Your goal is to find out which one of the provided offers is the cheapest and overall best fit for our needs. Read the [Instructions] and then use the [Input] as the comparison outline.”

Generally, the instructions consisted of instructions for the AI how to tone, structure, and present its answer. The input itself was a separate section that gave the final instructions (quantities, unit prices, hourly rates etc.) for the GenAI to conduct the bid comparison successfully. Figure 13 showcases the first part of the input which clearly emphasizes the objective for the GenAI tool to achieve.

[Input]

You are a procurement engineer in a construction company with 10 years of experience procuring subcontractor services for complex construction projects. Your task is to conduct a bid comparison based on the provided **six (6)** offer PDFs to assist procurement professionals. The goal is to deliver a structured analysis of the offers and recommend the most cost-effective option.

Figure 13. The inputs first initial prompt.

The prompt, consisting of the instructions and input, was structured so that every section would complement each other and provide the best possible output from the GenAI tool.

4.2.6 Testing the Prompt for Bid Comparison – Case Study

The testing phase was initiated by linking the RFQ and six bids from social module cleaning providers to the GenAI-tool. This type of procurement was selected for testing due to its relatively simple nature. It would not have been advisable to begin the research with complex procurements, as it is more reasonable to start with basic use cases and evaluate what kind of results can be achieved.

The procurement in question involved the cleaning of temporary social and office modules such as break areas, restrooms, and changing rooms used by the site workers and personnel on an office building construction site. In addition, the cleaning service included replenishment of hygiene products during the service time. Figure 14 below illustrates the real manual comparison results, where subcontractor B was identified as the most cost-effective option based on pricing, cleaning frequency, and the scope of the services provided based on the RFQ.

[illegible]

Figure 14. Manually conducted bid comparison for cleaning services.

This procurement represents a project-specific subcontracting procurement as categorized by (Junnonen & Kankainen, 2012). The cleaning services are delivered directly on-site and are closely integrated with the daily operations and schedule of the construction project. The cleanliness of the modules has a direct impact on the productivity and well-being of the workers and personnel, which is why this procurement can be categorized as such.

The GenAI tool was given an expected output to strive towards for which consisted of providing a comprehensive bid comparison including every provided offer. The project specifications, below, are listed in the same manner as they were in the initial prompt.

1. Cleaning of social modules (two) times per week (7 modules)
2. Cleaning of office modules (one) times per week (14 modules)
3. Additional cleaning times for social modules (800 times)
4. Service time (118 weeks)

The 1st iteration of the prompt, which included the above specifications, was pasted to the GenAI tools text box. The first answer ever provided by the AI-tool was invalid. The calculations were off, and the tool failed to offer any kind of analysis of the linked PDF files. Figure 15 below shows the first response given by the GenAI-tool.

Expected Output						
Provide a calculated comparison including every provided offer for the specified duration and cleaning requirements:						
Contractor	Office Module Cleaning (€)	Social Module Cleaning (€)	Hourly Rate (€)	Total Cost (€)	Contingency (10%) (€)	Final Cost (€)
A	17	20	N/A	TBD	TBD	TBD
B	5	6	N/A	TBD	TBD	TBD
C	Phase 1: 9 / Phase 2: 8	Phase 1: 7 / Phase 2: 6.20	Additional work: 28	TBD	TBD	TBD
D	8.50	9.40	28	TBD	TBD	TBD
E	7.10 + VAT	7.50 + VAT	Additional work: 27.50 + VAT	TBD	TBD	TBD
F	54	63	N/A	TBD	TBD	TBD
Note: The final cost calculations will be completed once all unit prices are aggregated based on the expected frequency of services over the project duration.						
By following these steps, we can ensure a thorough evaluation of each offer, leading to an informed decision that optimizes both cost and service quality for our construction project at construction site X.						

Figure 15. First answer provided by the GenAI-tool during testing phase.

The answer made it clear that remodifications to the prompt needed to be made. The future versions of the prompt were modified and tested with clearness and end result in mind. It was essential that the AI would recognize what it needed to do and in what kind of output it should present its answer.

After re-engineering the prompt by modifying both the instructions and the structure of the input to be more concise and directive in tone, another testing phase was initiated with a promising outcome. The total costs for the different subcontractors were almost identical compared to the manually conducted comparison, especially A, B, and C. This can be seen from Figure 16 below.



Figure 16. Bid comparison total costs after re-engineering the prompt.

It was noted that even with the same initial data (prompts, PDF files) GenAI tool offers different answer every time a new chat is generated. The first three costs from A to C were right but the variation becomes visible after this since D, E, and F are wrong. This means that the tool cannot be trusted, and it needs supervision. Figure 17 below showcases how the generated answer from the GenAI-tool is different with the same initial data as used in Figure 16. From the two, Figure 16 results are closer to the manually conducted comparison.

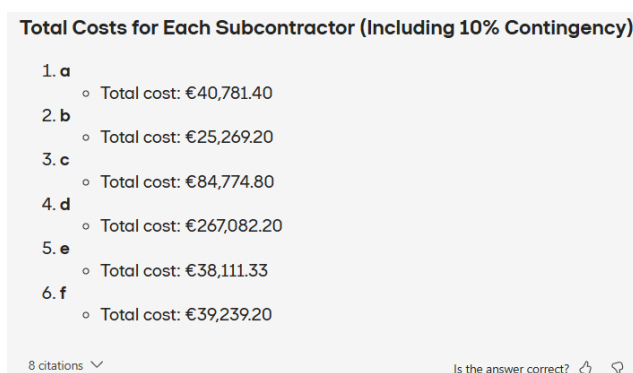


Figure 17. Bid comparison total costs through a new chat.

Table 6 below summarizes the variations between the total costs showcased in Figures 16 and 17.

Table 6. Variation of answers provided by the GenAI-tool in separate chats with same initial data.

Option	Figure 16	Figure 17	Variation
A	40 781 €	40 781,40 €	0,4 €
B	25 269 €	26 269,20 €	0,2 €
C	84 774 €	84 774,80 €	0,8 €
D	267 854 €	267 082,20€	771,8 €
E	29 757 €	38 111,33 €	8354,33 €
F	33 761 €	39 239,20 €	5478,2 €

Because of the differences between the provided answers, more variations of different prompt layouts were engineered and tested. It became obvious that three out of six individual offers contained too much information for the GenAI-tool to handle. This is why a simpler approach was tested. This meant adding only three offers and the RFQ at the same time and letting the AI generate an answer based on them.

Pasting three offers from A, B, and C at the same time provided almost the same total costs as the manually conducted comparison. In real life the costs for A, B, and C were 40 800€, 25 269 €, and 84 836€. This is due to simpler layout of the offers which only showed €/module cleaning prices for social- and office modules. Figure 18 below shows the AI's presentation of the calculations and the final costs between subcontractors A, B, and C with option B being the only offer that the GenAI-tool calculated correctly.

Total Cost Calculation (118 weeks)	
Subcontractor A	
1. Office modules: 14 modules * €8.50 * 118 weeks = €14,028	
2. Social modules: 7 modules * €9.40 * 2 times/week * 118 weeks = €15,512	
3. Additional cleaning times: 800 times * €9.40 = €7,520	
4. Total cost before contingency: €14,028 + €15,512 + €7,520 = €37,060	
5. Total cost with contingency (10%): €37,060 * 1.10 = €40,766	
Subcontractor B	
1. Office modules: 14 modules * €5 * 118 weeks = €8,260	
2. Social modules: 7 modules * €6 * 2 times/week * 118 weeks = €9,912	
3. Additional cleaning times: 800 times * €6 = €4,800	
4. Total cost before contingency: €8,260 + €9,912 + €4,800 = €22,972	
5. Total cost with contingency (10%): €22,972 * 1.10 = €25,269	
Subcontractor C	
1. Office modules: 14 modules * €17 * 118 weeks = €28,028	
2. Social modules: 7 modules * €20 * 2 times/week * 118 weeks = €33,040	
3. Additional cleaning times: 800 times * €20 = €16,000	
4. Total cost before contingency: €28,028 + €33,040 + €16,000 = €77,068	
5. Total cost with contingency (10%): €77,068 * 1.10 = €84,775	

Figure 18. GenAI based bid comparison between three subcontractors.

After this, the remaining offers were tried to be linked one by one to the existing chat, but this turned out to provide wrong answers same as before.

It is important to acknowledge that offers D, E, and F were challenging to interpret even for a human. These offers contained extra costs and showcased services that were not even asked in the RFQ. This could possibly explain why the calculated costs varied so much in the GenAI tool compared to the manually conducted one. This further showcases why concise and uniform bid offers are important for the company.

In conclusion, offers A, B, C were the only offers that the AI could calculate in a structured form and almost correctly, with B being correct most of the time. The best outcome was achieved when only the three most simple bids were provided to the GenAI tool. This gave a perception that the GenAI-tool, combined with a good prompt provides rather reliable results if the offer

layout is simple. It is important to acknowledge that the summarized analysis generated by the AI, discarding the wrong calculations, was impeccably correct at the end of every comparison. Service provider B, as in real life, was deemed always to be the cheapest option by the AI. Figure 19 below shows cases that the analysis from a testing session was correct even though the output and calculations were off.

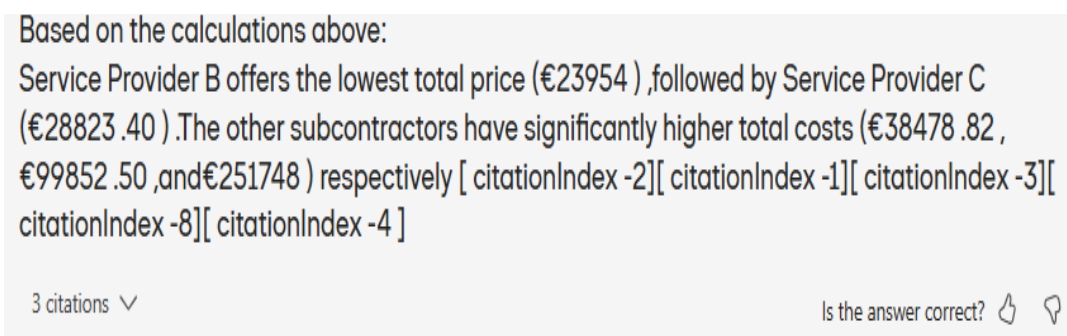


Figure 19. Option B being the cheapest service provider with otherwise failed comparison.

The performance of the GenAI tool was also analysed through the five different performance indicators mentioned before. The findings are presented in table 7 next page.

Table 7. Performance indicators and the initial findings based on them

Indicator	Explanation	Findings based on the testing
Time savings	AI should be able to reduce manual workloads and accelerate procurement processes.	AI has the potential to reduce manual workloads, and it did in the testing phase. The manually performed comparison took 30 minutes to complete, and the GenAI-tool took 2 minutes since all there needed to do was paste the prompt and the PDF-files. Still, the unreliability of the tool contradicts if it saves time rather than takes more time to check for errors. Currently the tool isn't reliable enough to be used for time-saving purposes.
Improved bid comparability	AI needs to correctly identify differences between bids and present clear comparisons.	AI didn't meet the expectations regarding this indicator. The presentations were mostly off, and the comparisons rarely were right except in the cases of A, B, C. Naturally, GenAI based bid comparison needs development work in the future.
Decision making support capabilities	If an AI-based tool provides structured reports that allow procurement teams to make informed decisions, it is doing what it is expected from it.	AI presented the generated answers always in a textual form, which subcontractor would be the cheapest, and every time it was right. The potential to use GenAI as an analysis tool is high and should be taken seriously.
Usability	Procurement professionals should start perceiving AI as an asset rather than an additional burden.	The usability of the GenAI tool was easy. There were no difficulties regarding using the tool. The platform and usability are not identified as a problem in a possible integration scenario.
Value addition	It was stated that AI's success can also be measured by whether procurement teams feel they have more time for strategic planning and value adding tasks rather than for repetitive work.	With a 100% success rate doing bid comparison, procurement teams would have substantially more time for other tasks. Currently, based on the testing, the time would be spent on checking for errors in the GenAI tool's answers.

4.3 Summary of Results

The results indicate that AI-based tools, particularly those utilizing GenAI, have notable potential to enhance procurement processes in construction. The most promising applications include 1) procurement package formation, 2) bid comparison, 3) contract oversight, and 4) supplier evaluation. GenAI-based tools hold strong potential, especially in improving bid comparability, highlighting inconsistencies, and supporting faster decision making. Successful integration of AI-based tools to company processes requires piloting before full on implementation. Procurement personnel will be more open to adopt new technologies when it is piloted and trained with data before integrating it to Skanska's processes. To measure the success rate of integrating AI-based tools to procurement processes five indicators were identified: 1) time savings, 2) improved bid comparability, 3) decision making support capabilities, 4) usability, and 5) value addition.

The testing of Skanska's own GenAI tool to conduct bid comparisons resulted in promising performance and potential for future usage. The GenAI tool provides relatively accurate bid comparisons when using well-structured and straightforward offers. With more complex offers, the tool offers inconsistent outcomes. At the time of this research, the GenAI tool is not capable of conducting an accurate bid comparison compared to a manually performed one. Additionally, repeated tests with identical inputs results in different outputs. This highlights inconsistency with current GenAI-based tools which yields for advancements in the technology field and development work in the future.

While AI is unlikely to fully automate procurement, it has a key role in potentially reducing errors, enhancing efficiency, and enabling procurement teams to focus on strategic decision-making. At this current stage, the GenAI tool doesn't provide time-savings reliably due to its inaccuracies while conducting bid comparisons.

Despite these numerical inaccuracies in the comparisons, the tool showed consistent accuracy in textual analysis, showcasing strength in summarizing and interpreting data contextually. This is the first step in utilizing GenAI-based tools in the procurement process and gradually develop the tool to conduct bid comparisons.

5 Discussion

This research aimed to identify opportunities for utilizing artificial intelligence in different phases of the procurement process and to explore its possible integration into them. The identified phases and the expected improvements through the integration of AI directly support this goal.

Possible integration strategies for AI into procurement processes by identifying prejudices and concerns were also explored. The results regarding discussions on piloting AI tools before integration, conservative culture of construction industry, and the need for systematic training address these prejudices and concerns.

This research had also a goal to develop and test a customized GenAI prompt within Skanska's own AI system against to a manually conducted one to enhance bid comparison's accuracy and efficiency. The case study provided insights into the performance of the tool and its potential for future usage. These findings support the development and testing goals.

Each one of the three research questions set for this research were answered, and relevant results were obtained:

What opportunities does artificial intelligence (AI) in general offer in different phases of the procurement process?

- **What kind of improvements can be achieved through its utilization to productivity, cost, and timelines?**

The interviews identified four key procurement phases where AI could provide the most value in: 1) bid comparison, 2) supplier evaluation, 3) procurement package formation, 4) contract oversight. These identified phases directly answer the set research question by pinpointing specific areas for AI integration at Skanska Talonrakennus Oy. The results indicate that AI can be

expected to boost productivity, cut costs, and shorten timelines through automatizing existing manually conducted processes.

What potential challenges and biases need to be identified and managed when integrating AI into procurement workflows, without disrupting existing processes during the integration phase?

The conservatism and resistance to changes in the construction industry are addressed as challenges when integrating new solutions to existing processes. The results from both the interviews and the case study discuss the necessity of piloting AI tools before full implementation, the importance of human expertise in decision making, and the need for quality procurement data. Also, the importance of cohesive training and education programs are important since they offer a more fluid transition to new processes. These points address the challenges and biases associated with AI integration and how it can be made as smooth as possible.

How can generative artificial intelligence (GenAI) be utilized to improve the accuracy and efficiency of bid comparison in procurement?

The testing phase of the research showed promising performance of Skanska's GenAI tool in bid comparisons, especially with well-structured offers. Skanska should look into the possibility of demanding suppliers to fill their bids to a standardized bidding template because this would increase the accuracy of GenAI-based comparisons.

Despite inconsistencies with the results, the tool showed efficient analysis structuring capabilities based on the bids. The obtained results were not as accurate as the manually performed one which indicates the need for human conducted comparisons. These findings align with the goal of improving bid comparisons accuracy and efficiency.

The most notable result obtained from this research is the obvious need for AI-based solutions to help procurement personnel to increase their productivity and have more time on strategic planning and value adding tasks. At this stage this can be achieved by using GenAI as an analysis tool when conducting bid comparisons. Other important factor to consider is that a GenAI-based bid comparison is fully possible and promising results were obtained.

It is valuable to note that the GenAI, at its current state, doesn't have the capability to generate accurate calculations since its development hasn't reached that mark yet. Hopefully in the future, as new innovations surface and new updates are released, this problem will be solved. This would mean that the usability and reliability of AI-based tools would rise, which in return promises better productivity for companies using them.

The implementation of GenAI-based tool to conduct bid comparison is not researched in existing literature as broadly as in this research. This may have to do with construction industry's conservative views or alternatively, the development work is happening behind closed doors. Either way, this research has initiated exploration in this area and supplies concrete data on the potential implementation of such solutions in the future.

In conclusion, development work is recalled for in the future and investment from Skanska Talonrakennus Oy regarding this research is required. This is because AI based technologies are gaining popularity, and in order to keep up with its progress, Skanska Talonrakennus Oy needs to invest in AI-based research.

5.1 Results Compared to Existing Research

Most notable finding in this research was the GenAI tools ability to perform textual analyses, which was already explored in previous literature by (Nyqvist et al., 2024b). Naturally, this thesis enforces GenAI's capability to

perform such tasks and is another stepping stone in validating its capabilities to the industry.

A surprising finding in this research was that the GenAI-tool could not, not even once, calculate same outcomes in the comparisons in separate chats with same initial data and prompt. This suggests unreliability in its current capabilities. This is indicated briefly through the studies conducted by (Abioye et al., 2021; Torro, 2024) where it is stated that AI's reliability is a problem since not even the creators of AI models don't know how the models come up with certain answers. Also, a study conducted by (Peltokorpi et al., 2025) states that GenAI-based tools suffer from inaccurate information and poor quality which correlates to this finding. The performance of GenAI-tools outputs with identical initial data in separate chats was not exclusively researched but rather implicated in existing literature, which is why this finding was surprising. One could think that same initial data and prompt would always result in same answers, since everything stays the same.

Compared to previous studies such as (Abioye et al., 2021), which highlights the general potential of AI in procurement processes but lacks concrete case-study of a GenAI-based bid comparison, this thesis provides new empirical insights into GenAI's performance with real bid data. Still, the generalizability of this research remains limited. The study focused on a single company and a specific type of procurement and procurement phase. Although the identified challenges such as bid complexity and AI's inconsistencies are likely to be present in future research, similar strengths such as textual summarization can be expected. Different performance of the GenAI-tool can be expected when testing it in different procurement process phases. For example, there is a possibility that GenAI-tool would potentially perform better when researching its capability to administrate real contracts.

6 Conclusions

The main goal of this thesis was to identify opportunities, potential improvements, and challenges when utilizing artificial intelligence in different phases of the procurement process and to explore its possible integration into them. Additionally, the performance of Skanska's own GenAI-based tool to perform a concrete bid comparison was tested with the goal of improving its accuracy and efficiency. The research had multiple goals which were complemented with the research questions revolving around them.

The results obtained from this study revealed four key procurement phases where AI could potentially improve the procurement processes at Skanska Talonrakennus Oy. These improvements are expected, if implemented, to boost productivity, reduce costs, and shorten timelines within the company. In addition to these findings, the ability of the GenAI-tool to perform textual analyses of the bids was seen as the first step for GenAI integration at Skanska Talonrakennus Oy's bid comparison process.

The research was constructed with an introduction to the topic including research questions and the scope of the study. After this, existing literature was researched in order to recognize current studies regarding the topic and views that are not yet researched. A two-part empirical section of this thesis aimed to answer the set research questions and this was achieved through two main methods. First, semi-structured interviews were conducted with an emphasis on all the research questions. After this, a case-study was conducted with the goal of concretely measuring GenAI-based tool's performance to make bid comparison more accurate and efficient. The success of the comparison was measured with identified indicators and a real comparison.

In conclusion, this research successfully answers the goals set for it and helps Skanska Talonrakennus Oy to develop its AI-based tools and procurement processes. The empirical research of this thesis explains the current status of AI-based tools in bid comparison and lays the foundation for future research.

The results obtained from this research were acknowledged at Skanska Talonrakennus Oy and observations were made. The GenAI's tools inaccuracy when conducting bid comparison was expected. At the moment, the tool isn't developed enough to successfully see through mathematical tasks with a 100% success rate, since there is always a risk for black box approaches. Reasoning models for ChatGPT were mentioned as being more mathematical oriented options which could help with the calculations being right more consistently. AI's potential to automate work processes was deemed important and this study's results are one step closer in achieving this goal. Collaboration between developers, construction professionals and AI experts is necessary to create more reliable AI applications in the future.

6.1 Future Overview

The development of AI, especially GenAI, has accelerated significantly in recent years by improving its accuracy and enhancing the ability to analyse longer and more complex texts. These advancements have undoubtedly increased its potential to support the automation of work processes, such as bid comparison, in the future.

The research identified possible future development opportunities that should be explored to fully harness AI-based tool's full potential.

1. First development opportunity is to start looking into possibilities on how AI could enhance the four phases where it was deemed to possibly provide value in. In addition to this, training and education programs

are needed to mitigate the biases regarding adapting new solutions in the company.

2. The ability of the GenAI tool to compare complex offers more accurately is a subject to look more into. The GenAI tool's inconsistency with complex offers in bid comparison suggests the need for further development and research. Future research should focus on improving the tool's ability to handle diverse and detailed bid formats. In addition, the developed prompt should naturally be refined along the improvement process, since GenAI tool's performance is heavily dependent on the clarity and simplicity of the initial prompt and input documents.

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Appendix 1 – Interview questions

Background questions

1. Briefly, what is your previous working experience in the construction field?
2. Have you done procurement for projects?
 - If so, what did you do?

Part 1

1. Can you describe the current procurement process at Skanska Talonrakennus Oy?
2. What are the most time-consuming tasks in procurement?
3. What challenges do you face while doing procurement for a specific construction project?
4. Have you encountered or heard about AI applications in procurement?
 - If so, what are they?
5. Which phases of the procurement process do you think AI could improve the most?
6. In your opinion, what would be the biggest benefits of integrating AI into procurement processes?
 - For example, cost savings, efficiency, accuracy
7. Are there any specific procurement tasks that you believe AI could automate or enhance?
 - For example, bid comparison? How?
8. What concerns do you have regarding the adoption of AI in procurement?
 - For example, reliability, integration
9. Do you foresee any biases in AI-driven procurement decisions? How do you think these biases could be mitigated?
10. What factors would make you more comfortable with using AI in procurement processes?
 - For example, step by step integration, wide introduction?

Part 2.

1. How are bids currently analyzed and compared at Skanska Talonrakennus Oy?
2. What are the main challenges in evaluating and comparing bids manually?
3. What criteria are most important when comparing bids?
4. Have you had any experience with AI-assisted bid comparison?
 - If so, what was your impression? If not, would you see it possible?
5. What risks or challenges do you see in using AI for bid analysis?
6. What level of human oversight do you think is necessary when AI is used for bid comparison at the very start of its integration?
7. What kind of information should an AI system consider when comparing bids (context, examples etc.)?
8. How would you measure the success of an GenAI-based bid comparison tool?

Final question

1. Anything to add regarding AI in procurement or bid comparison?